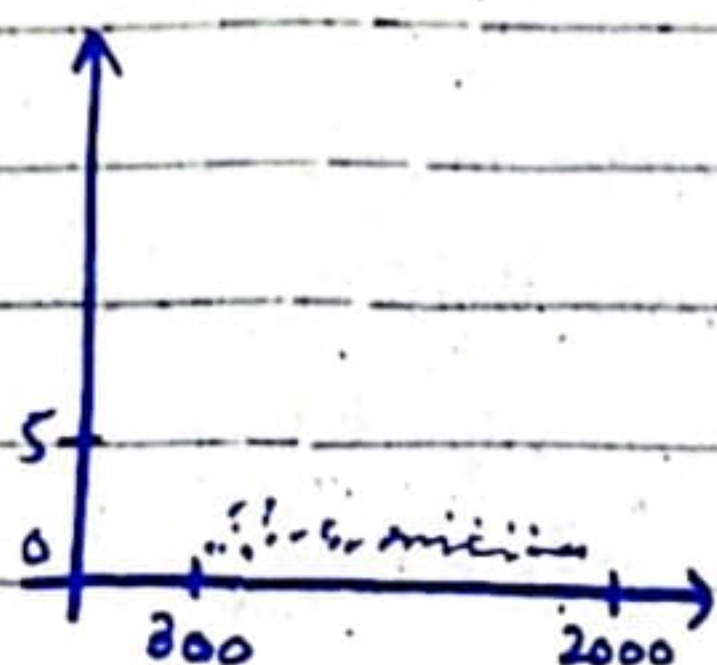


Feature scaling (Part 2)

How do you actually scale features?



$$300 < x_1 < 2000$$

$$0 < x_2 < 5$$

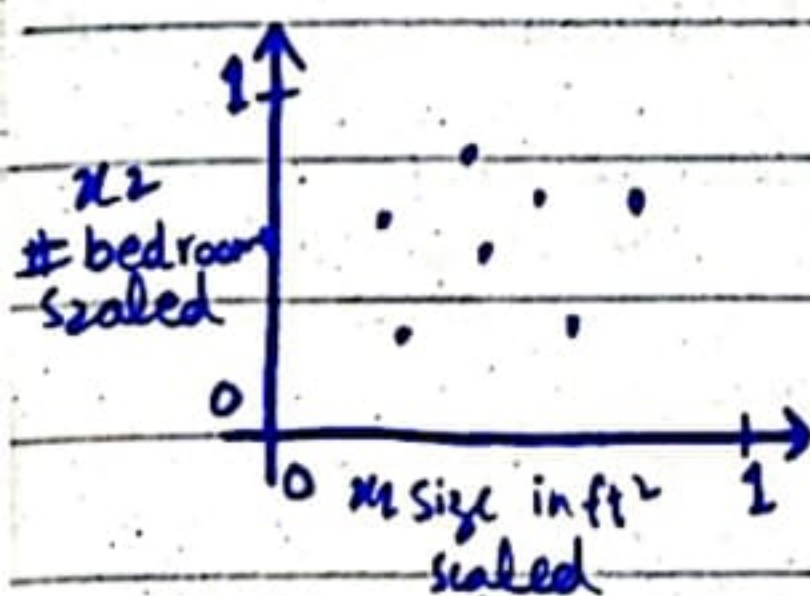
$$x_1 \text{ scaled} = \frac{x_1}{2000(\text{max})}$$

$$0.15 \leq x_1 \text{ scaled} \leq 1$$

$$0 \leq x_2 \text{ scaled} \leq 1$$

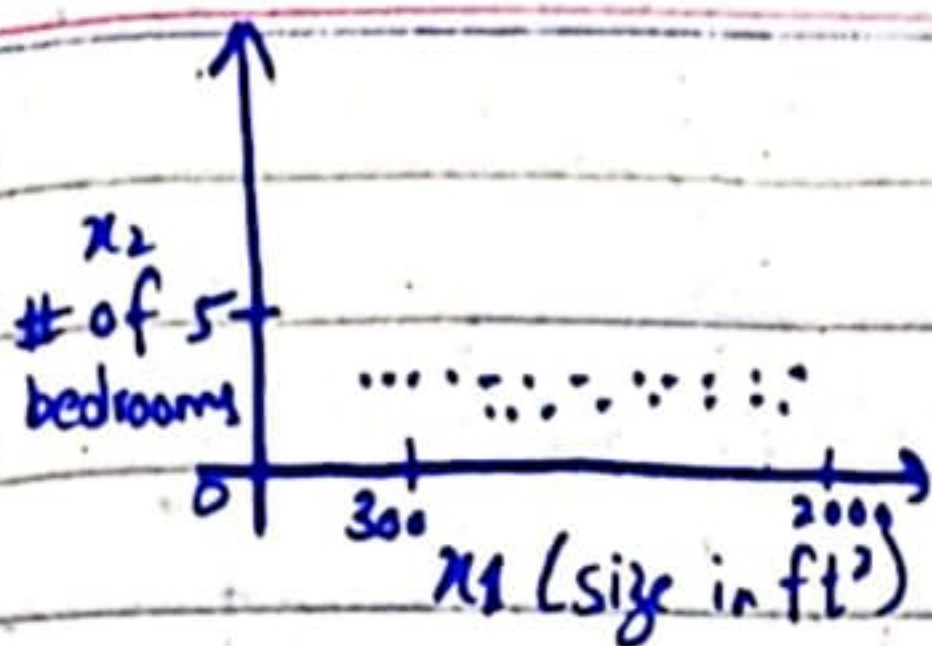
$$\frac{x_2}{5}$$

Now plotting will look like this:



In addition to the upper one, you can also do "normalization".

↳ Mean normalization



To calculate mean normalization of x_1
find avg (mean of x_1) on your training set.

call this mean (μ_1).
i.e. you may find mean is:

$$\mu_1 = 600$$

$$x_1 = \frac{x_1 - \mu_1}{2000 - 300} \rightarrow \text{max-min}$$

→ applied to every single house value.

$$\text{smallest } x_1 = 300$$

$$\rightarrow \text{normalized } x_1 = -0.18$$

$$\text{largest } x_1 = 2000$$

$$\rightarrow \text{normalized } x_1 = 0.82$$

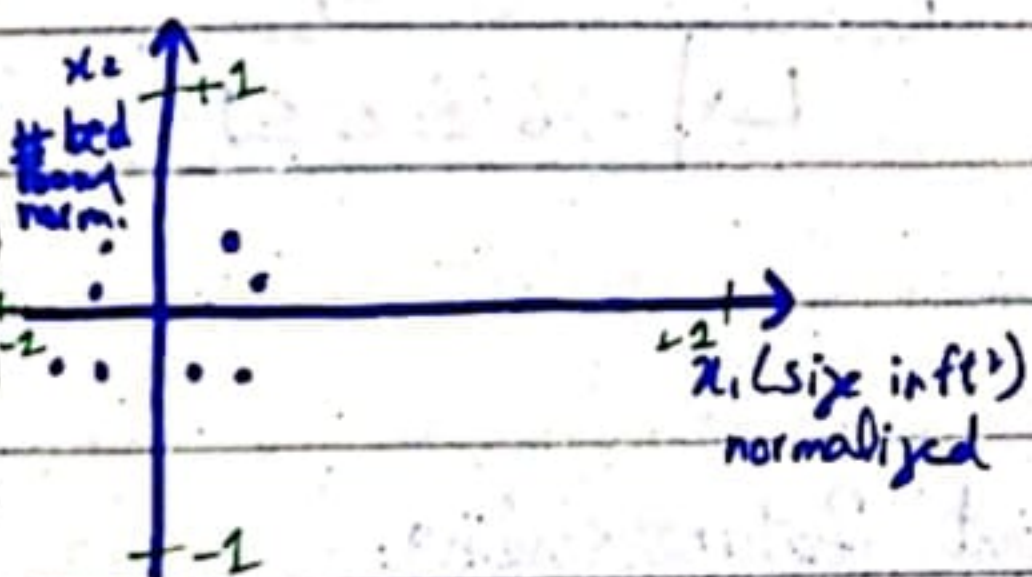
$$\text{so } -0.18 \leq x_1 \leq 0.82$$

Similarly for x_2 :

$$x_2 = \frac{x_2 - \mu_2}{5 - 0}$$

$$\rightarrow -0.46 \leq x_2 \leq 0.54$$

You start with original features and scale them so both of them are centered around zero.



whereas they only had values greater than zero now they have both negative & positive values.

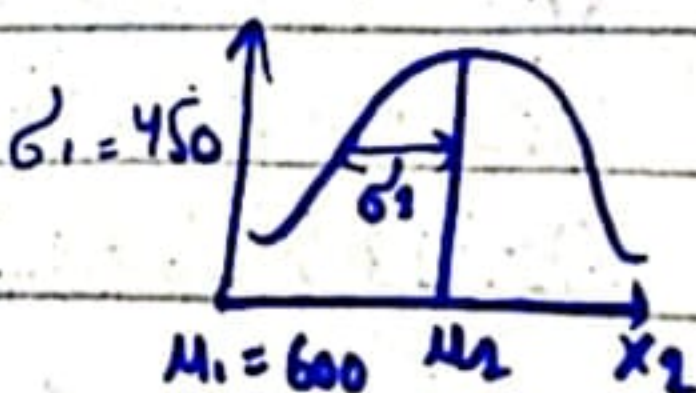
(usually may be b/w -1 & +1)

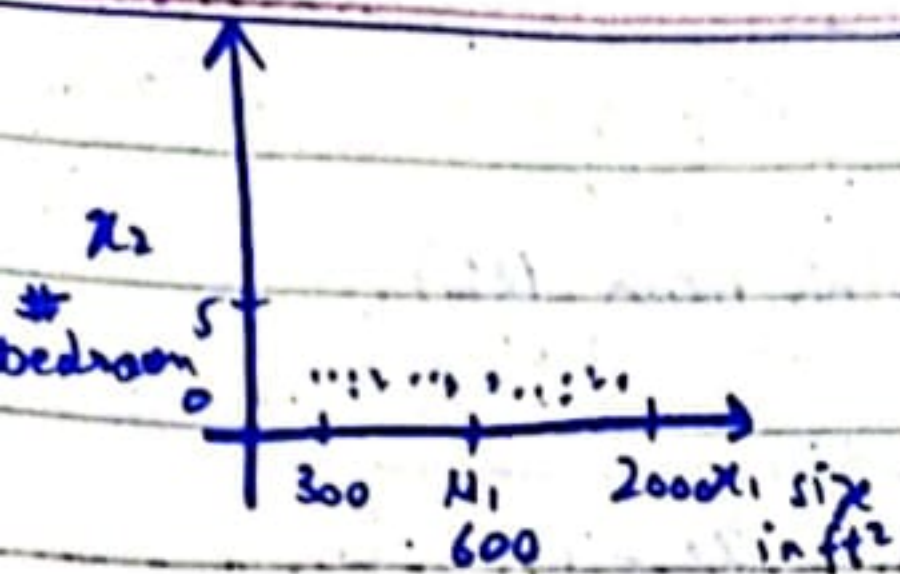
Third & last method:

"Z-score Normalization"

For this you need to calculate

"standard deviation σ "





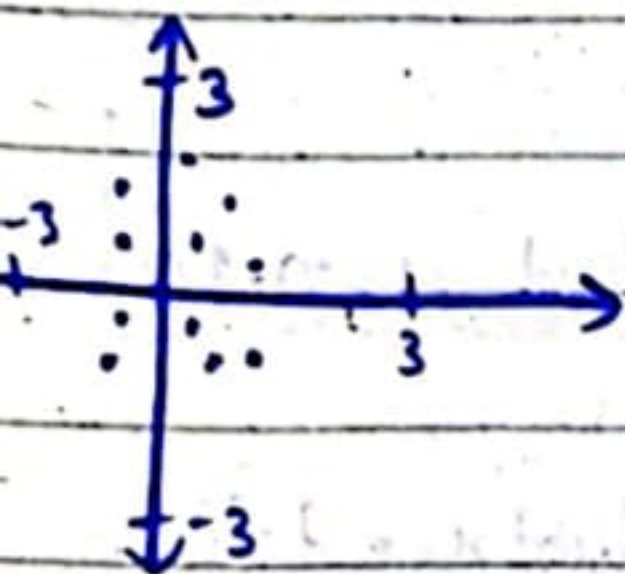
$$300 \leq x_1 \leq 2000$$



$$x_1 = \frac{x_1 - M_1}{s_1}$$

s_1

$$\rightarrow -0.67 \leq x_1 \leq 3.1$$



$$0 \leq x_2 \leq 5$$

$$x_2 = \frac{x_2 - M_2}{s_2}$$

s_2

$$\rightarrow -1.6 \leq x_2 \leq 1.9$$

For rule of thumb about feature scaling:

• aim for about $-1 \leq x_j \leq 1$ for each feature x_j .

even if $-3 \leq x_j \leq 3$ } acceptable ranges
 $-0.3 \leq x_j \leq 0.3$

1) $0 \leq x_1 \leq 3$

okay } can rescale but no harm in not

2) $-2 \leq x_2 \leq 0.5$

okay, } doing for both

but

3) $-100 \leq x_3 \leq 100$ too large $\rightarrow$ rescale

4) $-0.001 \leq x_4 \leq 0.001$ too small $\rightarrow$ rescale

5) $98.6 \leq x_5 \leq 105$ too large $\rightarrow$ rescale.

